

MyDesign Portfolio

Element H

Element H: Prototype Testing and Data Collection Plan - Mic Flags

Testing Details (H1, H2, H3, H4)

	requirement	testing setup and procedure	number of trials	expected range of results	pass/fail criteria, goal
1	The pictures on the mic flag do not fade High Priority	- use the image you plan on putting on the mic flag - print the image out twice - place both images on separate panels and cover them with a layer of tape - place one image in direct sunlight for 24 hours - have one without direct sunlight as a control panel - compare the images, ask 10 people whether the image appears different on the two panels	3 Tests, 1 control and 1 testing panel for each test	Image May Be: - Unchanged (0-1 People see change) - Slightly Faded (2-4 People see change) - Majorly Faded (5-10 People see change)	Fail: Logo Is Majorly Faded Pass: Logo is Slightly Faded Goal: Logo is Unchanged
2	Size of logo must be 75mm by 50 mm High Priority	- print out the panel - take a digital multimeter and find dimensions of the panel.	3 trials	Logo is larger than 75 mm by 50 mm Logo is thinner than 75 mm Logo is shorter than 50 mm Logo is shorter & thinner than 50 mm and 75 mm	Fail: Logo is thinner and shorter than requirements Pass: Logo is thinner or shorter than requirements Goal: Logo is taller and wider than requirements

3	The mic flags should be durable High Priority	- Print out flag - Drop flag from a height of 1 ft - Inspect the flag for any damages	3 drops	Flag suffers no damage Flags suffers minor scuffing that does not affect function Logo suffers major damage that affects function	Fail: Flag is no longer usable Pass: Flag suffers minor damage and is still usable Goal: Flag suffers no damage
4	Mic flags should be replaceable when damaged Moderate Priority	- Start timer - Slice the 3D model of the flag - Send the sliced flag to a 3D printer and begin print - Once the print is finished glue foam into the hole for the mic holder - Once the foam is glued in, stop the timer	3 prints	0s<Recording<48hrs	Fail: mean of the trials ≥ 24 hrs Pass: mean of the trials < 24 hrs Goal: mean of the trials < 12 hrs
5	Mic Flag must be able to fit on multiple types of mic holders Moderate Priority	- Find 3 different mic holders of different diameters - Place flag on holder - Shake holder for 1 minutes - Record any movement of flag from initial position (Use before and after pictures)	3 trials for each holder	Mic flag falls off holder Mic flag shifts on holder Mic flag does not move on holder	Fail: Mic Flag does not move on 1 holder Pass: Mic flag does not move on 2 holders Goal: Mic flag does not move on 3 holders

6	Corners of the mic flag must not be sharp Low Priority	- Take mic flag and a piece of cardboard - Drag flag on cardboard - Record damage made by flag on cardboard	3 trials	Mic Flag cuts into surface of cardboard Mic Flag leaves light mark on surface of the cardboard Mic Flag does not leave any mark on cardboard	Fail: Mic flag cuts into cardboard Pass: Mic flag leaves slight mark on cardboard Goal: Mic flag leaves no mark on flag
7	mass of prototype <100g Low Priority	- Take a digital scale. - Place the prototype on the scale. - Wait for the reading on the scale to stabilize before recording the value.	test cases: 3 # trials: 3	0kg<reading<1kg	Fail: mean of the trials $\geq 100g$ Pass: mean of the trials < 100g Goal: mean of the trials < 75g

Feedback From Field Experts (H5)

Jack S. - TV Production Student

1. Can't see logos anymore if pictures fade, will prevent mic flags from communicating the brand on it.
2. If smaller, the viewers will be unable to see the logo
3. If the mic stays intact while falling, it can be used in the future.
4. Very necessary as it will allow us to keep a constant supply of flags
5. If we were to use older holders, it will be ideal if they are compatible.
6. There is no need to risk poking an eye out
7. If it was too heavy, it would remove mobility of the mic

Overall, the requirements fit the needs of what we want. The order of priority follows the order that we care about each requirement. I agree with what ones are marked with each priority level.